

Calculation of the Static Forces Among the Permanent Magnets in a Halbach Array

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This paper concerns the analysis of the interaction force among the permanent magnets (PM) in a Halbach array as used in a moving-magnet planar actuator. The goal is to analyze the magnitude of these forces and to what extent the mechanical construction of an aluminium plate, on which the permanent magnets are mounted, is deformed by the interaction force among the permanent magnets. The interaction forces are obtained from analytical models which have been verified with an electromagnetic Finite Element Method (FEM) simulation. These results are used in a mechanical FEM simulation to determine the deformation of the magnet assembly.

Index Terms—Deformation of mechanical structure, Halbach array, magnetic force.

I. INTRODUCTION

IN recent years, magnetically levitated planar actuators [1], [2] have been developed as an alternative to xy -drives constructed of stacked linear actuators in demanding applications such as wafer stages. Moving-magnet planar actuators show promising performance, however, they do not yet offer the required positioning accuracy. In order to achieve a higher positioning accuracy, several steps have to be made. One of these steps is to model the flexible modes of the translator, which consists of an aluminium plate on which a Halbach array of permanent magnets is mounted. To this end, insight in the interaction force among the permanent magnets in the Halbach array is needed.

Previous research on the interaction forces between two permanent magnets has been carried out by among others [3] (concerning two parallel magnetized magnets) and [4] (concerning two orthogonally magnetized magnets). This paper uses the analytical models presented in [3] and [4] to calculate the static forces among the permanent magnets in a Halbach array as presented in Fig. 1. The force models are verified by means of electromagnetic FEM and the results are used to calculate the deformation of an aluminium plate on which the permanent magnets are mounted with mechanical FEM simulations.

II. DEFINITION OF THE HALBACH ARRAY

The geometry of the Halbach array analyzed in this paper is similar to the Halbach array presented in [1] which consisted of 11×11 permanent magnets alternatively magnetized in the z -direction. The Halbach array under study consists of 2×2 permanent magnets magnetized in the z -direction and is shown in Fig. 1. The Neodymium Iron Boron (NdFeB) permanent

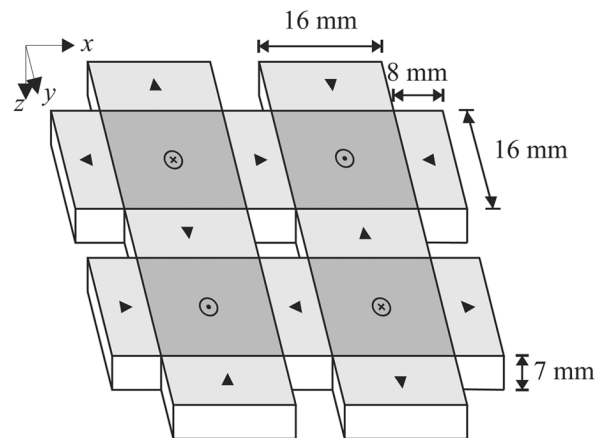


Fig. 1. 2×2 Halbach array of permanent magnets. The direction of magnetization for each magnet is indicated by the arrows.

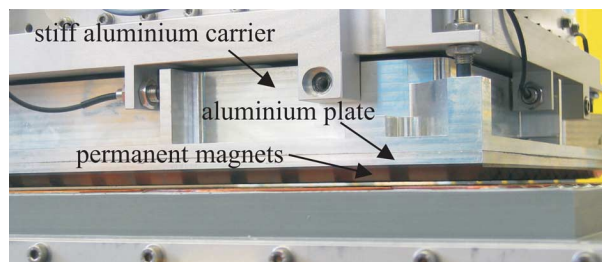


Fig. 2. Picture of the magnet plate mounted on an aluminium carrier as used in the setup presented in [1].

magnets have a remanent flux density, B_r , equal to 1.24 T and a relative permeability, μ_r , equal to 1.04. It is assumed that the permanent magnets are positioned at a negligible distance from each other. The total size of the Halbach array consisting of 11×11 permanent magnets magnetized in the z -direction is 272 mm $\times$ 272 mm $\times$ 7 mm. A picture of the magnet plate mounted on an aluminium carrier as used in the setup presented in [1] is shown in Fig. 2.

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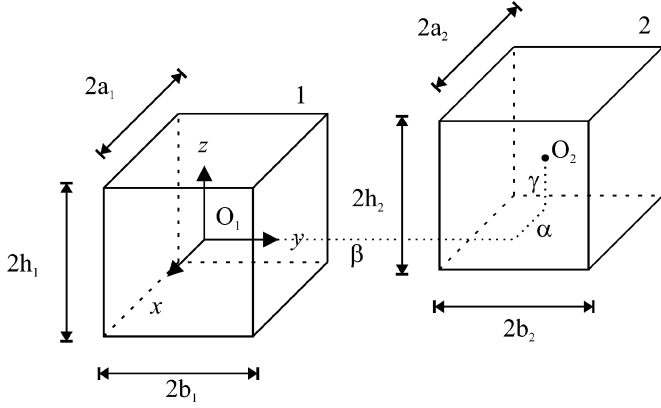


Fig. 3. Dimensions of two cuboidal permanent magnets. O_1 and O_2 are the center of permanent magnet 1 and 2 respectively. (α, β, γ) indicates the distance between O_1 and O_2 in the (x, y, z) -direction.

III. CALCULATION OF THE INTERACTION FORCE AMONG PERMANENT MAGNETS USING FEM

A FEM package such as Flux 3D [5] is commonly used to calculate the interaction forces in electromagnetic applications. These packages numerically calculate the fields in a system on each element in the mesh and the time required to solve the problem is therefore highly dependent on the mesh size. Since most of the electromagnetic FEM packages use the virtual work method to calculate the force, a layer of air surrounding the region of interest is needed. Hence, to calculate the force on individual permanent magnets, the permanent magnets have to be positioned at a certain distance between each other. Also, when creating a mesh for such a structure, the mesh size in the layer of air should be chosen equal to or smaller than the distance between two magnets to obtain an accurate model. This results in a trade-off between the distance between two permanent magnets and the number of mesh elements. In practice, for the Halbach array presented in [1], the distance between the magnets is approximately $50 \mu\text{m}$ to $100 \mu\text{m}$, which would require a very dense mesh. This makes FEM unsuitable for solving this specific problem.

IV. CALCULATION OF THE INTERACTION FORCE AMONG PERMANENT MAGNETS USING ANALYTICAL MODELS

For an ironless structure, the interaction force acting on each magnet can also be calculated using analytical models, based on the surface charge method, which are presented in [3] and [4]. These models assume a relative permeability $\mu_r = 1$, which will result in a slight error due to a difference compared to the actual value of $\mu_r = 1.04$. Using these analytical models, the interaction force can be calculated with zero distance between two permanent magnets. Consider the two permanent magnets as shown in Fig. 3 with two possible magnetization variations as shown in Fig. 4. For the two magnets shown in Fig. 4(a), the force between the magnets is given by [3]

$$F_{(x,y,z)} = \sum_{i=0}^1 \sum_{j=0}^1 \sum_{k=0}^1 \sum_{l=0}^1 \sum_{p=0}^1 \sum_{q=0}^1 (-1)^{i+j+k+l+p+q} \times \psi_{(x,y,z)}, \quad (1)$$

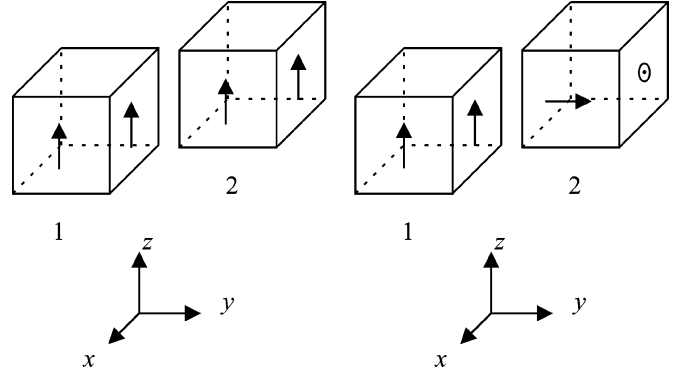


Fig. 4. (a) Two permanent magnets with parallel magnetization vectors. (b) Two permanent magnets with orthogonal magnetization vectors.

where

$$\psi_x = \frac{1}{2}(T^2 - U^2) \ln(R - S) + ST \ln(R - T) + TU \arctan\left(\frac{ST}{RU}\right) + \frac{1}{2}RS, \quad (2)$$

$$\psi_y = \frac{1}{2}(S^2 - U^2) \ln(R - T) + ST \ln(R - S) + SU \arctan\left(\frac{ST}{RU}\right) + \frac{1}{2}RT, \quad (3)$$

$$\psi_z = -SU \ln(R - S) - TU \ln(R - T) + ST \arctan\left(\frac{ST}{RU}\right) - RU. \quad (4)$$

For the two permanent magnets shown in Fig. 4(b), the force $F_{(x,y,z)}$ between the permanent magnets can be found with (1) where $\psi_{(x,y,z)}$ is equal to [4]

$$\psi_x = \frac{1}{2} \left[\left(\arctan\left(\frac{U}{S}\right) + \arctan\left(\frac{TU}{RS}\right) \right) S^2 + 2TS - 3US - 2T \ln(U + R)S - 2T^2 \arctan\left(\frac{S}{T}\right) + T^2 \arctan\left(\frac{SU}{TR}\right) + U \left(U \left(\arctan\left(\frac{S}{U}\right) + \arctan\left(\frac{ST}{UR}\right) \right) - 2T \ln(S + R) + 2S \ln(R - T) \right) \right], \quad (5)$$

$$\psi_y = \frac{1}{2} \left[U(R - 2S) - (S - T)(S + T) \ln(U + R) + 2S \left(T \left(\arctan\left(\frac{U}{T}\right) + \arctan\left(\frac{ST}{UR}\right) \right) + U \ln(R - S) \right) \right], \quad (6)$$

$$\psi_z = \frac{1}{2} \left[D(A - 2B) - (B - C)(B + C) \ln(D + A) + 2B \left(C \left(\arctan\left(\frac{D}{C}\right) + \arctan\left(\frac{BD}{CA}\right) \right) + D \ln(A - B) \right) \right]. \quad (7)$$

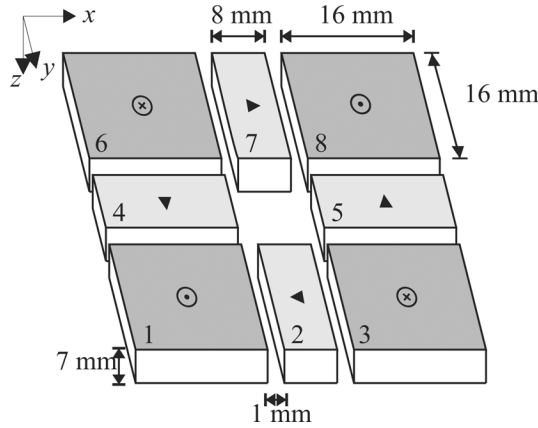


Fig. 5. Halbach array of permanent magnets as used to compare results between FEM and the analytical models, the numbers in the bottom left of each permanent magnet indicate the magnet number N .

The variables S, T, U, R, B, C, D and A are given by

$$\begin{aligned}
 S &= \alpha - (-1)^i a_1 + (-1)^j a_2, \\
 T &= \beta - (-1)^k b_1 + (-1)^l b_2, \\
 U &= \gamma - (-1)^p h_1 + (-1)^q h_2, \\
 R &= \sqrt{S^2 + T^2 + U^2}, \\
 B &= \alpha - (-1)^i a_1 + (-1)^j a_2, \\
 C &= -\gamma + (-1)^k h_1 - (-1)^l h_2, \\
 D &= \beta - (-1)^p b_1 + (-1)^q b_2, \\
 A &= \sqrt{B^2 + C^2 + D^2}.
 \end{aligned} \quad (8)$$

The force between any other combination of magnetization directions can be found with these equations by rotation of the coordinate system. Hence, the interaction forces among the magnets in a Halbach array as shown in Fig. 1 can be calculated by means of these equations.

To verify the consistency between FEM and the analytical models, the model as shown in Fig. 5 is implemented in Flux 3D with $\mu_r = 1.04$ and $\mu_r = 1$ and compared to results found from the analytical models. The model as shown in Fig. 5 has a distance of 1 mm between the magnets since, as explained earlier, this allows for decreasing the density of the mesh. The value of the force in the (x, y, z) -direction for permanent magnet with number $N = 1$ and $N = 2$ as numbered in Fig. 5 is shown in Table I. The results show that the values obtained from FEM with $\mu_r = 1$ correspond well to the analytical models. Hence, the error when using the analytical models is due to a difference in μ_r . For the most significant force component, F_z , the error is below 4%.

The analytical model is used to calculate the interaction forces acting on the magnets in the 2×2 array shown in Fig. 1 and the result is shown in Fig. 6, with the force (in N) in the (x, y, z) -direction beside the arrows. Note that the force is not indicated for each magnet and that the total force acting on the permanent magnet array equals zero. The z -component of the force acting on the permanent magnets magnetized in the z -direction is most significant and equal to approximately

TABLE I
FORCE VALUES FOUND FROM FLUX 3D AND THE ANALYTICAL MODELS

N		Flux 3d		analytical models
		$\mu_r = 1.04$	$\mu_r = 1$	$\mu_r = 1$
1	F_x	2.81 N	2.52 N	2.54 N
	F_y	2.81 N	2.52 N	2.54 N
	F_z	46.5 N	47.2 N	47.2 N
2	F_x	0.01 N	0.01 N	0.00 N
	F_y	-6.13 N	-6.40 N	-6.40 N
	F_z	-46.5 N	-47.2 N	-47.2 N

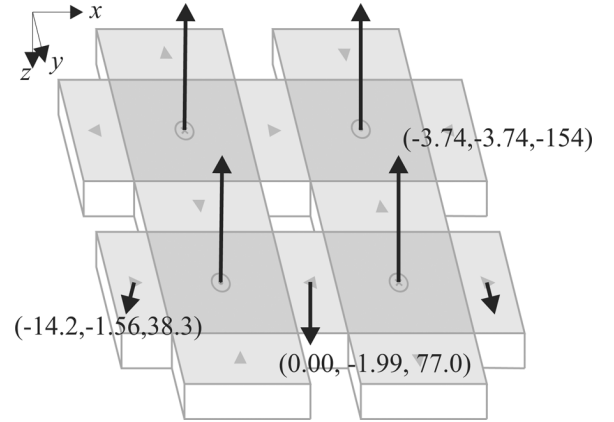


Fig. 6. Forces acting on the permanent magnets in the Halbach array, the numbers beside the vector indicate the force in the (x, y, z) -direction.

154 N. For a Halbach array consisting of 11×11 permanent magnets alternatively magnetized in the z -direction, this force component increases with less than 0.1%. Hence, the interaction force is independent of the number of magnets in the array.

V. CALCULATION OF THE DEFORMATION OF THE STRUCTURE

The mechanical structure on which the array of permanent magnets is mounted will deform due to the magnetic forces which the permanent magnets exert on each other. This deformation has been calculated by means of mechanical FEM simulations for which several assumptions and simplifications are made. First, the magnet plate on which the Halbach array is mounted is solid and made of isotropic aluminium with a Young's modulus of 60 GPa, a density of 2600 kg/m³ and a thickness of 5 mm. The aluminium carrier as shown in Fig. 2 is not considered in these simulations. Second, only the forces acting in the z -direction are considered, since these force components are dominant for the deformation, and it is assumed that these forces act on the surface of the magnet plate directly above the center of the permanent magnet. Third, because of the non-zero distance between the magnets and the low stiffness of the glue compared to aluminum, the total stiffness of the magnet assembly is dominated by the aluminum plate and the influence of the magnets and the glue is neglected. Fourth, the magnet plate is fixed at the corner points and the reaction force at these fixed boundaries is zero (it is verified that similar results are obtained for other boundary conditions). A schematic overview of a cross-section of the aluminium plate with the magnets is shown in Fig. 7. The size of the mesh elements in the (x, y, z) -direction is $2 \times 2 \times 1$ mm. Under these assumptions, the deformation of the magnet plate due to the magnetic forces

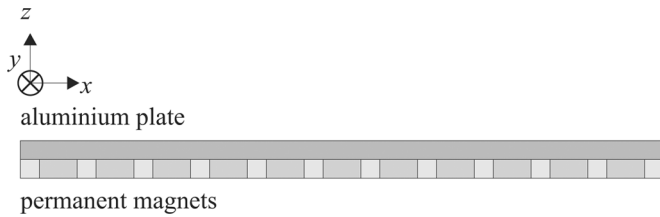
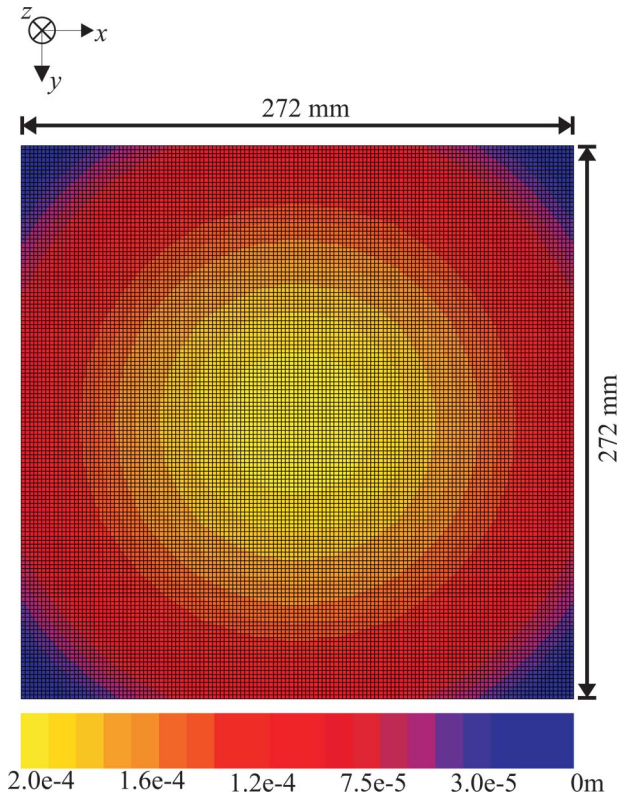


Fig. 7. Cross-section of the aluminium plate with the permanent magnets.

Fig. 8. Deformation of the magnet plate. The legend indicates local displacement in the z -direction.

generated by the Halbach array is simulated with Marc/Mentat [6]. The results are shown in Fig. 8.

The results show that the outer edges of the magnet plate tend to bend downwards by approximately 0.2 mm with respect to the center of the magnet plate. These results are similar to the data obtained from measurements on the magnet plate during manufacturing. This is significant considering the accuracy demanded by the applications for which these types of planar actuators are being designed. In practice, a 0.1 mm variation in the

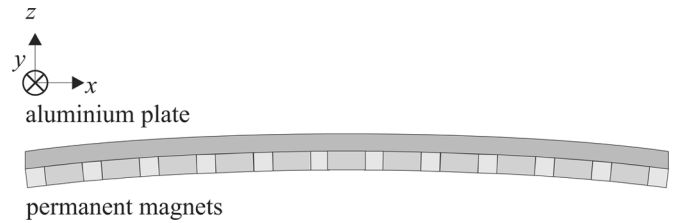


Fig. 9. Impression of a cross-section of the deformed aluminium plate with the permanent magnets.

airgap results in a 1.8% variation in the force production of the planar actuator presented in [1]. In the planar actuator setup, the aluminium plate is mounted on an additional aluminium carrier as shown in Fig. 2 which improves the stiffness of the structure. Hence, the deformation of this mechanical construction is smaller, however, the shape of the deformation is similar. An impression of a cross-section of the deformed aluminium plate with the permanent magnets is shown in Fig. 9.

VI. CONCLUSION

This paper concerns the interaction forces between the permanent magnets in a Halbach array. Electromagnetic FEM packages are unsuitable for these kind of force analyses. Instead, the forces are calculated by means of analytical models and the results show that the magnitudes of these forces are significant. These results are used to perform mechanical FEM simulations on a mechanical structure on which the Halbach array is mounted to determine to what extent the mechanical structure is deformed by the magnetic forces generated by the Halbach array. The results show that the structure is significantly deformed.

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